

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0655

Speed 138 kV
April 15, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0655. The Interconnection Customer proposes a 469.5 MW natural gas combined cycle facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on July 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 243 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,680,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DAMON to SILVER BAY POWER 138 kV line	SILVER BAY POWER to HOTEL RD 138 kV L	159.3	230	33.9
56TH STREET to OATES REIT 138 kV L	SPEED to SILVER BAY POWER 138 kV Line	156.8	506	100.0
56TH STREET SUBSTATION to SANTA MONICA 138 kV	SILVER BAY POWER to SPEED 138 kV L	151.4	544	100.0
SANTA MONICA to HAMBRICK 138 kV	SILVER AY POWER to SPEED 138 kV	148.6	566	100.0
BROCKPORT to NAROD 138 kV line	SILVER BAY POWER to SPEED 138 kV	147.8	545	100.0
BARNHAM to NAROD 138 kV	SILVER BAY POWER to SPEED 138 kV L	143.4	583	100.0
BARNHAM to GAGA 138 kV	SILVER BAY POWER to SPEED 138 kV line	141.0	605	100.0
SILVER BAY POWER to SPEED 138 kV line	GAGA to SPEED 138 kV L	138.9	626	100.0
SPEED to GAGA 138 kV	SILVER BAY POWER SUBSTATION to SPEED 138 kV	138.9	626	100.0
COULEE AVE to SPORN 345 kV	HANGING ROCK to INTERNATIONAL PAPER 500 kV	129.5	1,109	5.7
SILVER BAY POWER to HOTEL RD 138 kV	LIVERMORE to LEE COUNTY SOLID WASTE ENERGY SUBSTATION 345 kV	124.2	350	22.5
OSAWATOMIE to SOUTHSIDE WASTEWATER TREATMENT PLANT 100 kV	TOLEDO BEND to KILLEN 500 kV	122.6	143	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NARCOOSSEE to BRINCKEN'S CORNER 100 kV	NARCOOSSEE to LAWRENCE TAP 100 kV	122.2	142	6.6
JULIETTE to ALIBATES 138 kV	PINE BLUFF WATSON CHAPEL to CARRAIZO 1201 345 kV	121.9	368	27.0
CPU SILICON SUBSTATION to LANHAM 138 kV	LIVERMORE to LEE COUNTY SOLID WASTE ENERGY 345 kV	120.0	286	6.2
PICNIC POINT to NARCOOSSEE 100 kV Cir. 1	NARCOOSSEE to PICNIC PT 100 kV Cir. 2	119.3	179	6.4
NARCOOSSEE to PICNIC PT SUBSTATION 100 kV #2	NARCOOSSEE to PICNIC POINT 100 kV #1	119.3	179	6.4
ELDORADO to ACTION SUBSTATION 138 kV Line	PINE BLUFF WATSON CHAPEL to CARRAIZO 1201 345 kV	117.9	638	32.9
KINGSROW to CPU SILICON 138 kV	LEE COUNTY SOLID WASTE ENERGY to LIVERMORE 345 kV L	117.4	306	6.2
WAGON MOUND to UNIVERSIITY 138 kV	CARRAIZO 1201 to PINE BLUFF WATSON CHAPEL 345 kV	116.3	272	16.1
ALIBATES to ACTION 138 kV	CARRAIZO 1201 to PINE BLUFF WATSON CHAPEL 345 kV line	115.3	682	32.9
MARACAIBO to UNIVERSIITY 138 kV	PINE BLUFF WATSON CHAPEL to CARRAIZO 1201 345 kV line	115.1	255	11.9
PINE BLUFF WATSON CHAPEL to CARRAIZO 1201 345 kV	LEE COUNTY SOLID WASTE ENERGY to LIVERMORE 345 kV line	114.8	1,227	54.9
BINGEN to PRAGER 138 kV L	OSAWATOMIE to WAGNER LAKE 230 kV	114.0	249	10.9
SANTA ROSA BEACH to TULE 138 kV	DAVISVILLE to N APPLETON 138 kV	113.6	176	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DAMON to SILVER BAY POWER 138 kV line	Rebuild line with higher-capacity conductor	\$3,485,000	36% (\$1,249,000)
56TH STREET to OATES REIT 138 kV	Rebuild line with higher-capacity conductor	\$3,185,000	36% (\$1,142,000)
56TH STREET SUBSTATION to SANTA MONICA 138 kV	Rebuild line with higher-capacity conductor	\$4,557,000	36% (\$1,633,000)
HAMBRICK to SANTA MONICA 138 kV L	Rebuild line with higher-capacity conductor	\$4,279,000	36% (\$1,534,000)
BROCKPORT to NAROD 138 kV line	Rebuild line with higher-capacity conductor	\$2,921,000	36% (\$1,047,000)
BARNHAM to NAROD 138 kV	Rebuild line with higher-capacity conductor	\$20,974,000	36% (\$7,518,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GAGA to BARNHAM 138 kV	Rebuild line with higher-capacity conductor	\$1,346,000	36% (\$483,000)
SILVER BAY POWER to SPEED 138 kV	Rebuild line with higher-capacity conductor	\$14,917,000	36% (\$5,347,000)
SPEED to GAGA 138 kV	Rebuild line with higher-capacity conductor	\$2,728,000	36% (\$978,000)
COULEE AVE to SPORN 345 kV	Rebuild line with higher-capacity conductor	\$70,901,000	6% (\$4,570,000)
SILVER BAY POWER to HOTEL RD 138 kV	Rebuild line with higher-capacity conductor	\$14,429,000	23% (\$3,288,000)
OSAWATOMIE to SOUTHSIDE WASTEWATER TREATMENT PLANT 100 kV	Rebuild line with higher-capacity conductor	\$2,297,000	34% (\$781,000)
NARCOOSSEE to BRINCKEN'S CORNER 100 kV	Rebuild line with higher-capacity conductor	\$2,375,000	26% (\$613,000)
JULIETTE to ALIBATES 138 kV	Rebuild line with higher-capacity conductor	\$10,837,000	26% (\$2,780,000)
CPU SILICON to LANAM 138 kV Line	Rebuild line with higher-capacity conductor	\$2,947,000	10% (\$281,000)
PICNIC POINT to NARCOOSSEE 100 kV	Rebuild line with higher-capacity conductor	\$1,884,000	26% (\$493,000)
NARCOOSSEE to PICNIC PT SUBSTATION 100 kV #2	Rebuild line with higher-capacity conductor	\$1,047,000	26% (\$274,000)
ELDORADO to ACTION 138 kV	Rebuild line with higher-capacity conductor	\$3,992,000	22% (\$858,000)
KINGSROW to CPU SILICON 138 kV	Rebuild line with higher-capacity conductor	\$5,232,000	10% (\$498,000)
WAGON MOUND to UNIVERSITY 138 kV	Rebuild line with higher-capacity conductor	\$10,585,000	32% (\$3,340,000)
ALIBATES to ACTION 138 kV	Rebuild line with higher-capacity conductor	\$18,056,000	22% (\$3,881,000)
MARACAIBO to UNIVERSITY 138 kV	Rebuild line with higher-capacity conductor	\$8,409,000	32% (\$2,657,000)
PINE BLUFF WATSON CHAPEL to CARRAIZO 1201 345 kV	Rebuild line with higher-capacity conductor	\$49,072,000	22% (\$10,685,000)
BINGEN to PRAGER 138 kV L	Rebuild line with higher-capacity conductor	\$1,645,000	15% (\$253,000)
SANTA ROSA BEACH to TULE 138 kV line	Rebuild line with higher-capacity conductor	\$4,614,000	11% (\$497,000)
	Total allocated to project		\$56,680,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.